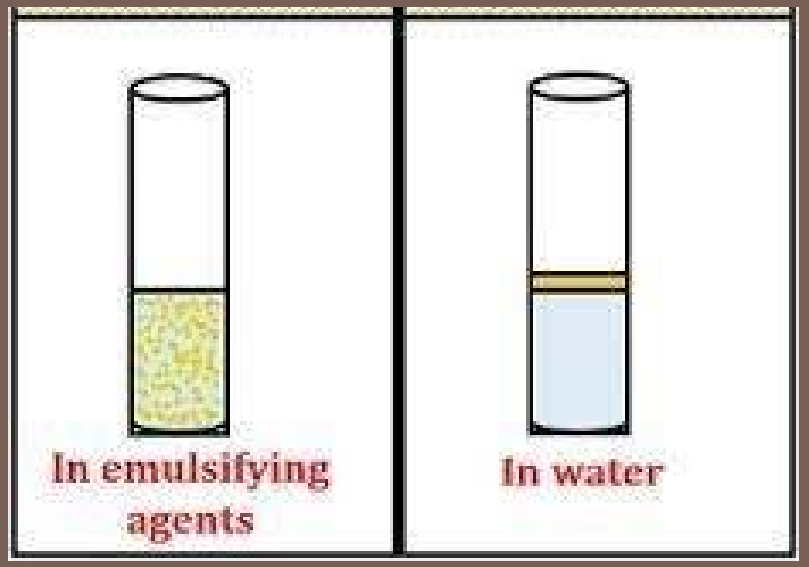
EMULSIFICATION TEST
&
HAY'S TEST



Dr. Humaira Aman Ali

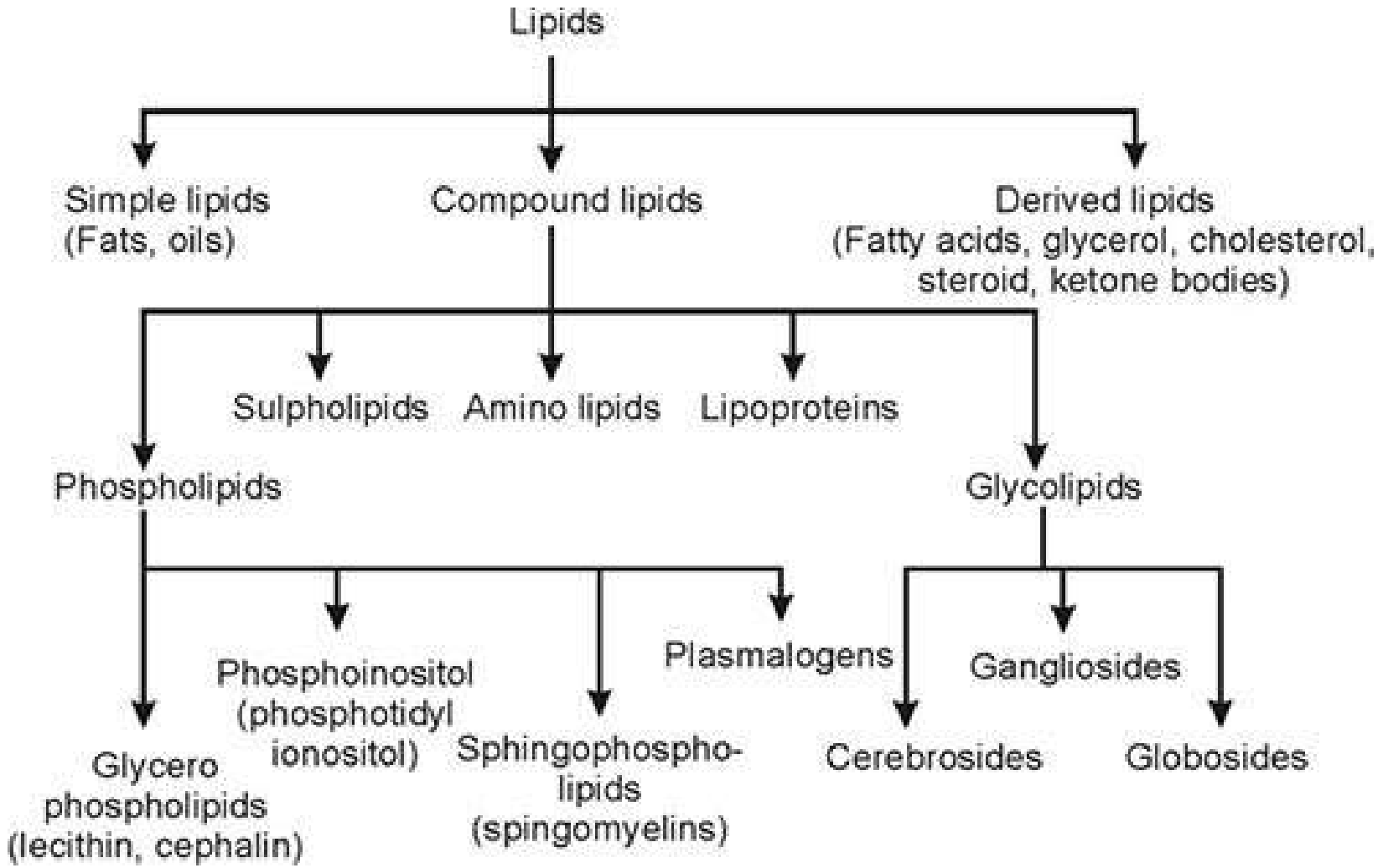
Introduction of Lipids

- The lipids are organic substances occurring in plant and animal tissues and belong to a very **heterogeneous group of substances** which have only a few properties in common.
- Lipids have been defined as substances having the following characteristics :-
- They are insoluble in water (hydrophobic) but soluble in non-polar solvents commonly termed fat solvents e:g ether , chloroform , benzene and acetone etc . This is because although certain lipids contain ionized groups (eg phosphate , serine and choline) , but the **bulk of any lipid is non-polar** .
- Their primary building blocks are fatty acids , glycerol , sphingosine and sterols .

Definition

- “Any of a class of organic compounds that are fatty acids or their derivatives and are insoluble in water but soluble in organic solvents”. .
- Essential structural component of living cells (along with proteins and carbohydrates), that helps shape, protect, and insulate the cell. It also provides an energy source..
- Lipids are a broad group of naturally-occurring molecules which includes fats and fats related substances, waxes, sterols, fat-soluble vitamins (such as vitamins A, D, E and K),natural oil , monoglycerides, diglycerides, phospholipids, and others.
- It contains hydrogen, carbon, and oxygen.

Classification of Lipids



Emulsification Test

- **Objective**: To detect the presence of lipid in the given sample.
- **Principle:**
  - The emulsification test is a commonly used method to determine the presence of lipids in any sample.
  - Emulsification of fats means breaking up of fat globules into much smaller emulsion droplets and make it stable by emulsifying agent.
  - Fat emulsification is important in the process of digestion in the intestine where different intestinal enzymes along with bile salts play their role.
  - Emulsifiers are substances that reduce surface tension at

Requirements

- Given sample of oil
- Distilled water
- Emulsifying agent
- Test tubes and pipette

Procedure

- Take 2 ml of water in a test tube A, add 2-4 drops of given oil sample and shake well for about a minute. Note the appearance of solution.

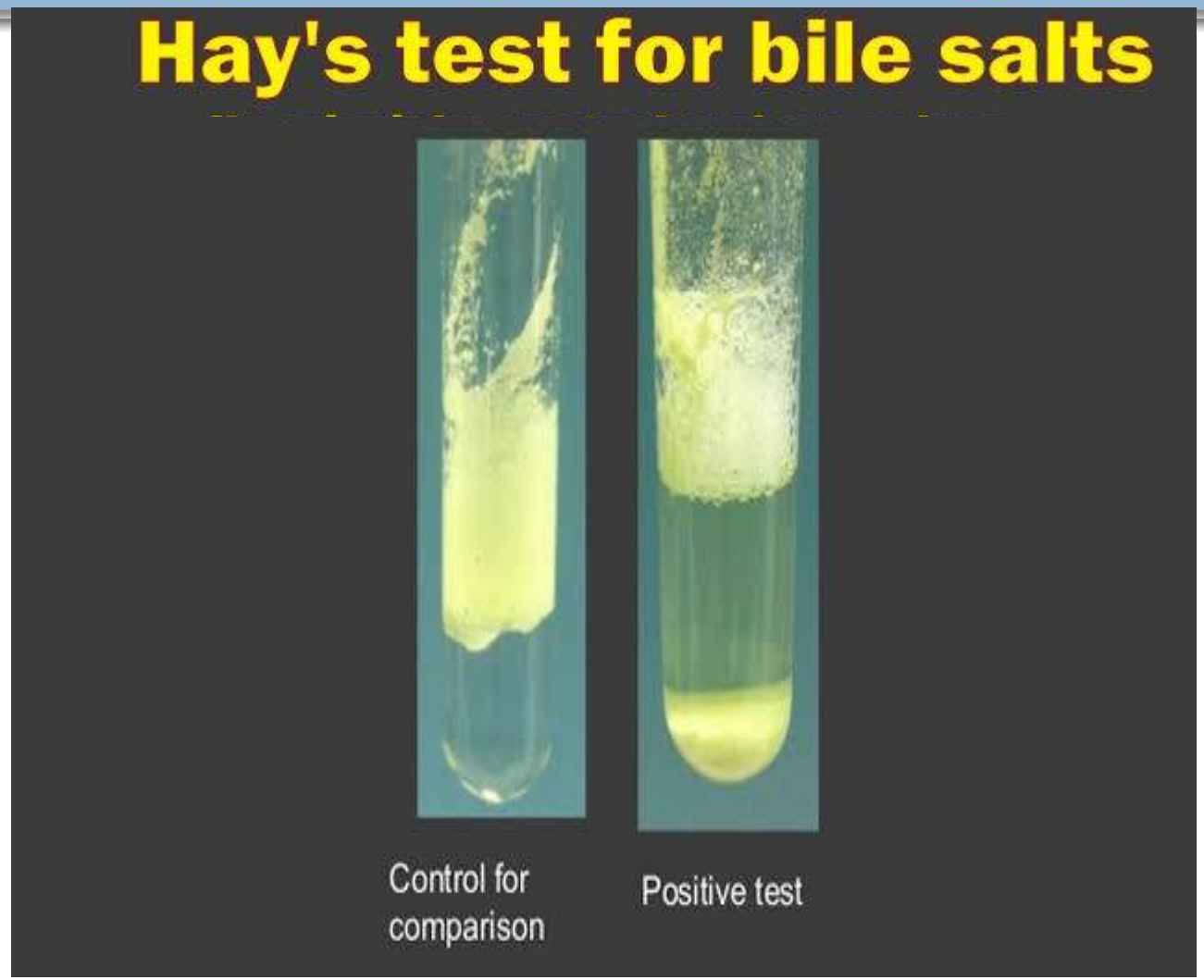
- Take 2 ml of sample with emulsifying agent in test tube B, add 2-4 drops of given oil sample and shake well for about a minute.

DETECTION OF BILE SALT (HAY'S TEST)

- **Object:** To detect bile salts in the given sample by Hay's test.
- **Principle:** When sulfur powder is added to bile salts, they lowers the surface tension and allow the sulfur powder to sink.
- **Requirement:**
  - Sample
  - Sulfur powder
  - Clean & dry test tubes
  - Pipettes

- **Procedure:**
  - Take 2 test tubes and mark them.
  - In first tube take 5 ml distilled water and sprinkle sulfur powder on surfaces.
  - In second tube take 5ml sample and sprinkle sulfur powder on surfaces.
- **Observation:**
  - In first tube powder remains on surface of water.
  - In second tube powder sink and settle at bottom.

Hay's test for bile salts



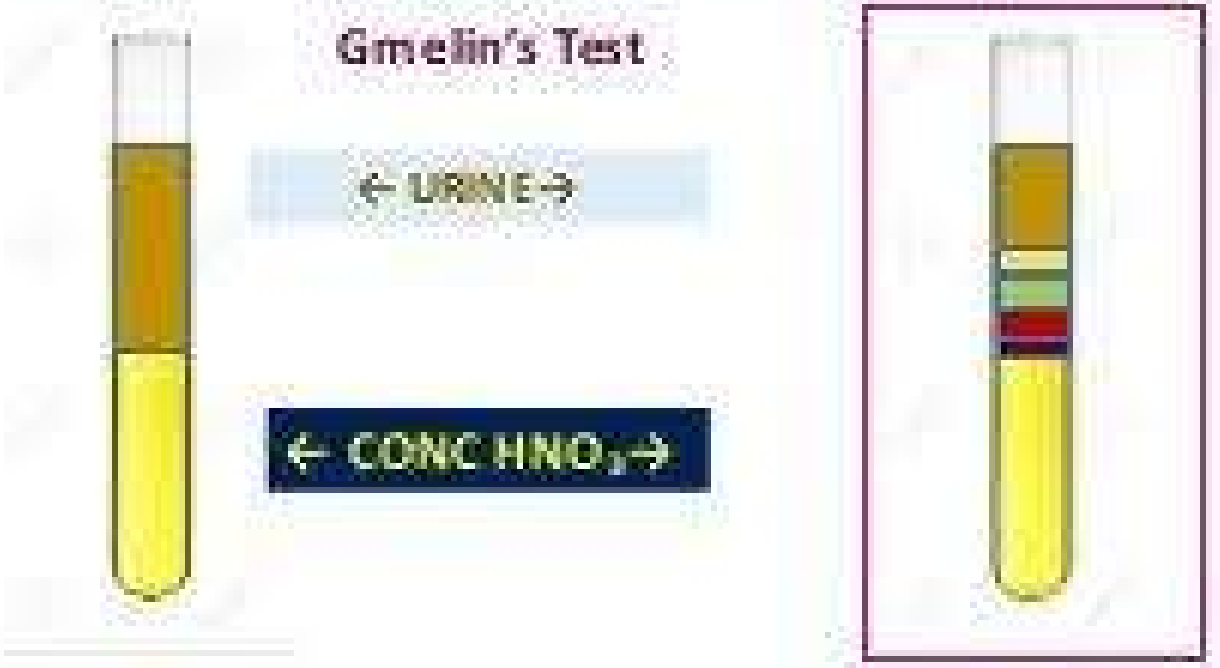
DETECTION OF BILE PIGMENTS

- **Object:** To detect bile pigment in the given sample by GMELIN'S TEST.
- **Principle:** Concentrated HNO_3 oxidizes bilirubin to various colored compounds namely biliverdin (green), bilicyanin (blue), bilifuschin (red) and choletelin (yellow).
- **Reuirements:**
  - Sample
  - Concentrated HNO3
  - Clean & dry test tubes
  - Pipettes

Procedure:

- Take 3 ml of conc. HNO_3 and add 3ml of sample along the walls of the tube to avoid mixing of two solutions. Multicolored rings (green, blue, red, and violet etc.) will be formed.

Gmelin's test in urine : Liver function test based upon bile pigment metabolism



Bilirubin –yellow ,Biliverdin –green ,Bilifuschin –Red , Bilicyanin –violet

THANK YOU